

Logical database design and the Relational Model

Modern Database Management

12th Edition

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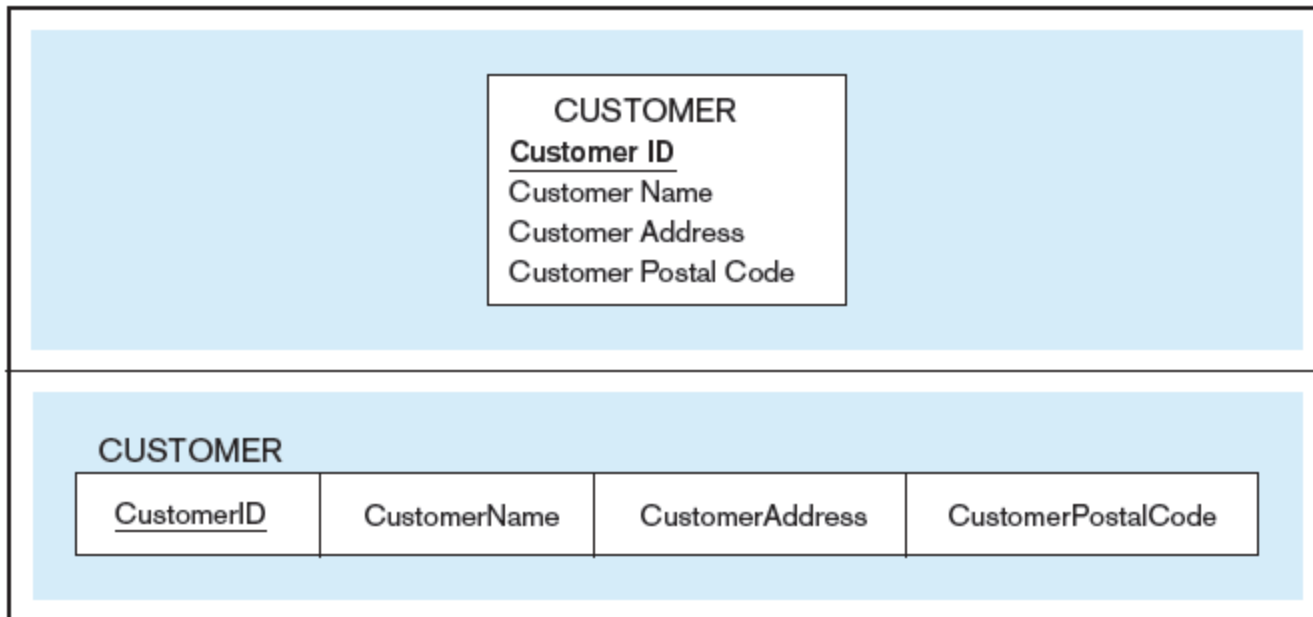
Lecture 11

Transforming EER diagrams into Relations

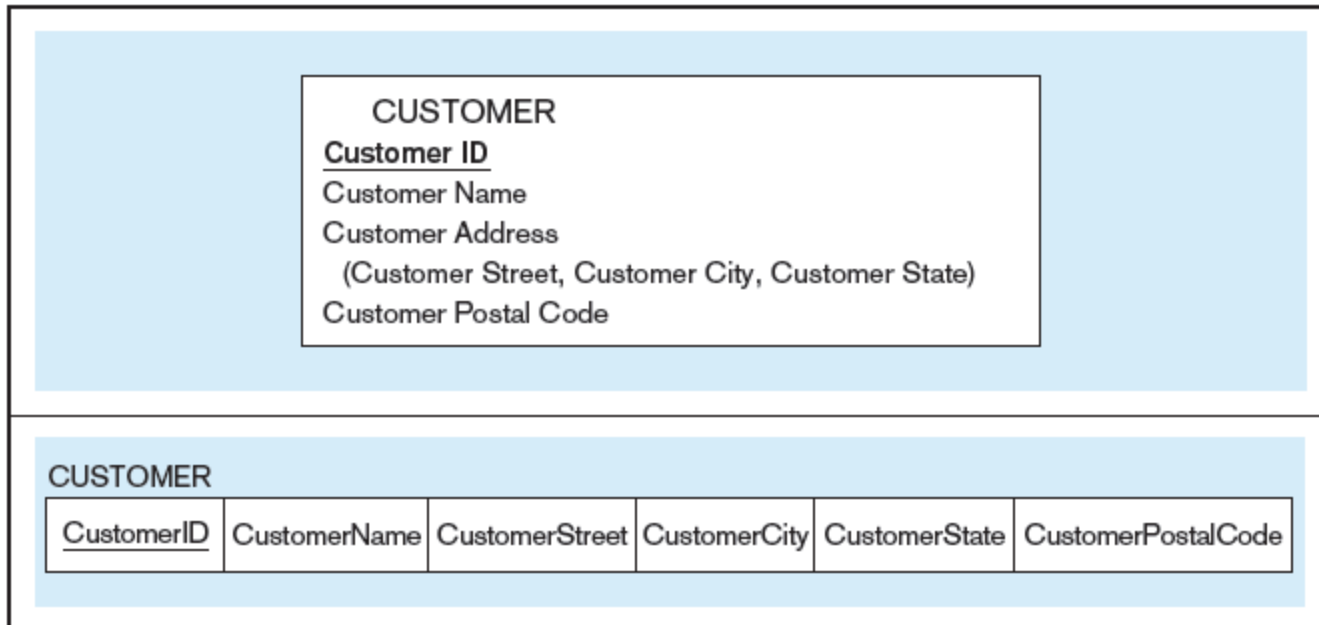
Map Regular Entities

- ❖ The name given to the relation is generally the same as the entity type
- ❖ Each simple attribute of the entity type becomes an attribute of the relation.
- ❖ The identifier of the entity type becomes the primary key of the corresponding relation.
- ❖ COMPOSITE ATTRIBUTES: only the simple components of the composite attribute are included in the new relation as its attributes.

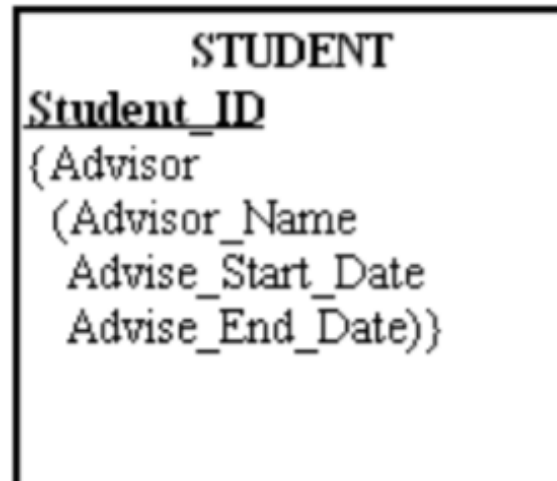
Map Regular Entities contd..



Mapping composite attributes



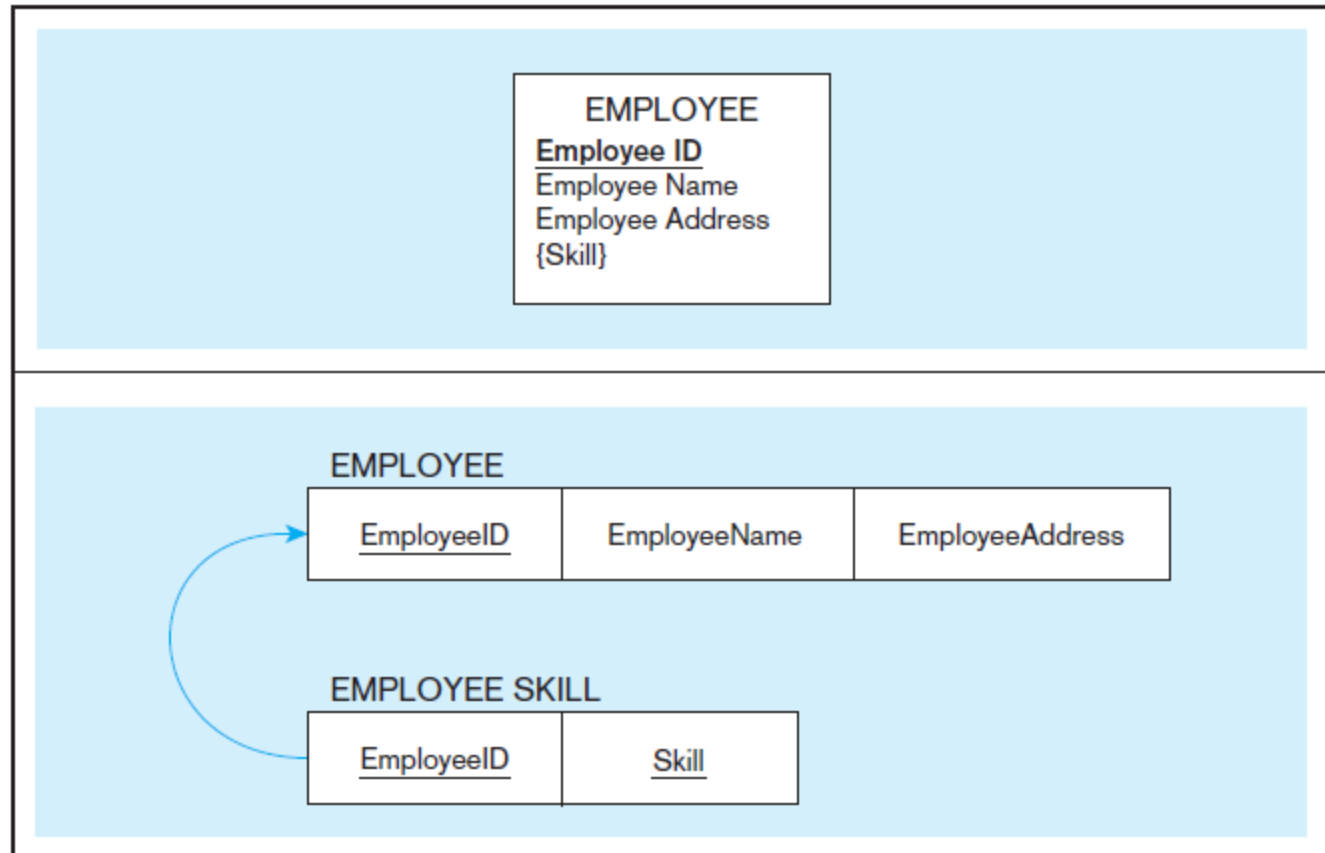
ACTIVITY-I



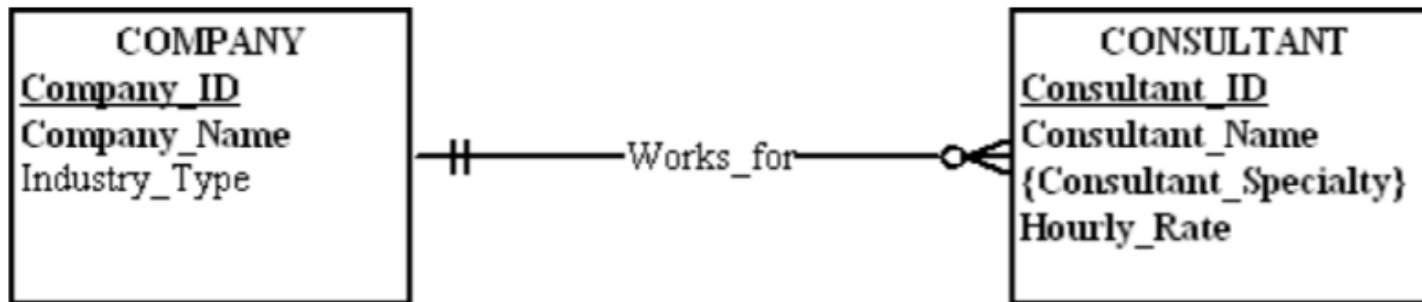
MULTIVALUED ATTRIBUTES

- ❖ The first relation contains all of the attributes of the entity type except the multivalued attribute
- ❖ The second relation contains two attributes that form the primary key of the second relation
- ❖ The first of these attributes is the primary key from the first relation, which becomes a foreign key in the second relation.
- ❖ The second is the multivalued attribute.
- ❖ The name of the second relation should capture the meaning of the multivalued attribute.

MULTIVALUED ATTRIBUTES contd..



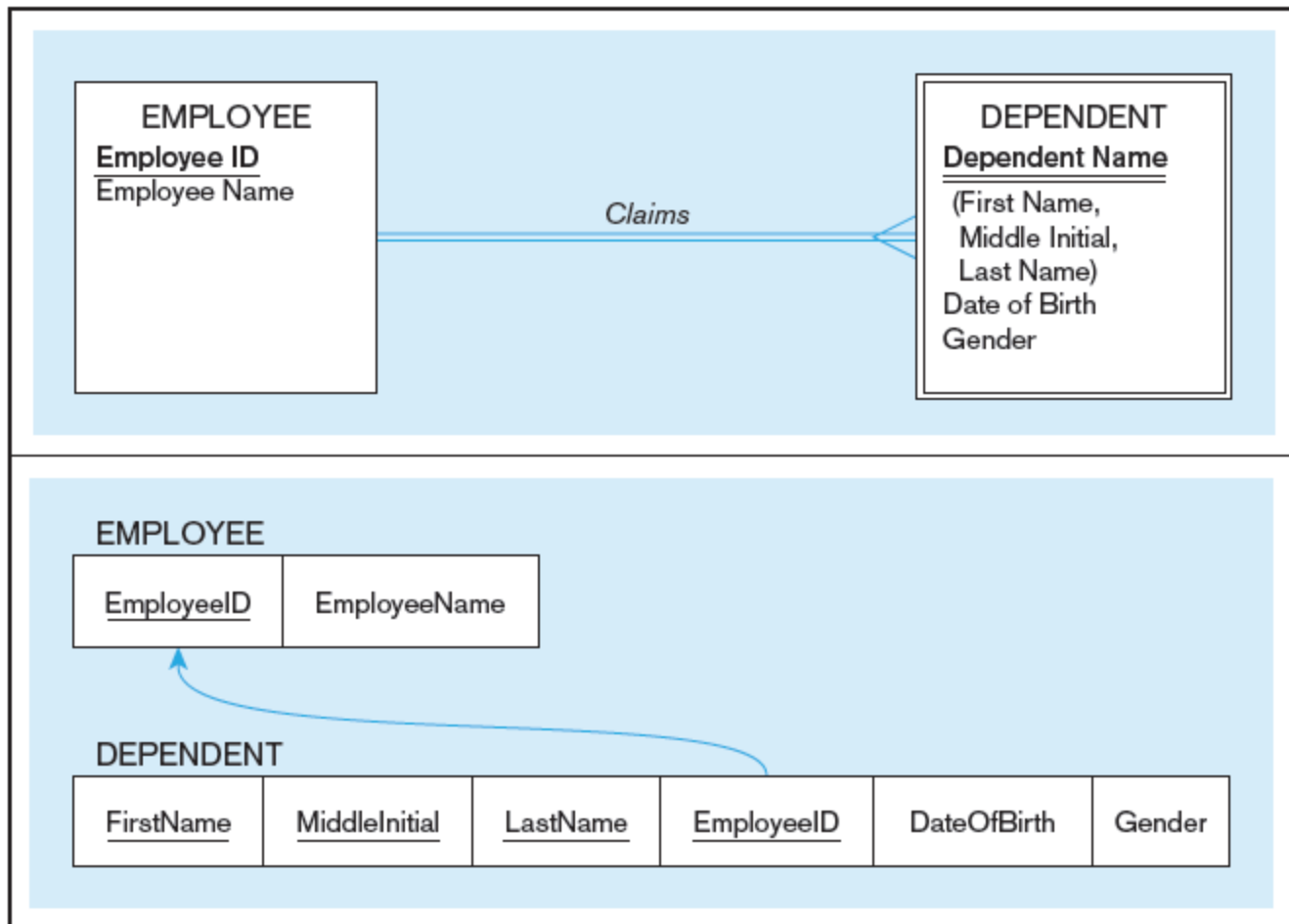
ACTIVITY-II



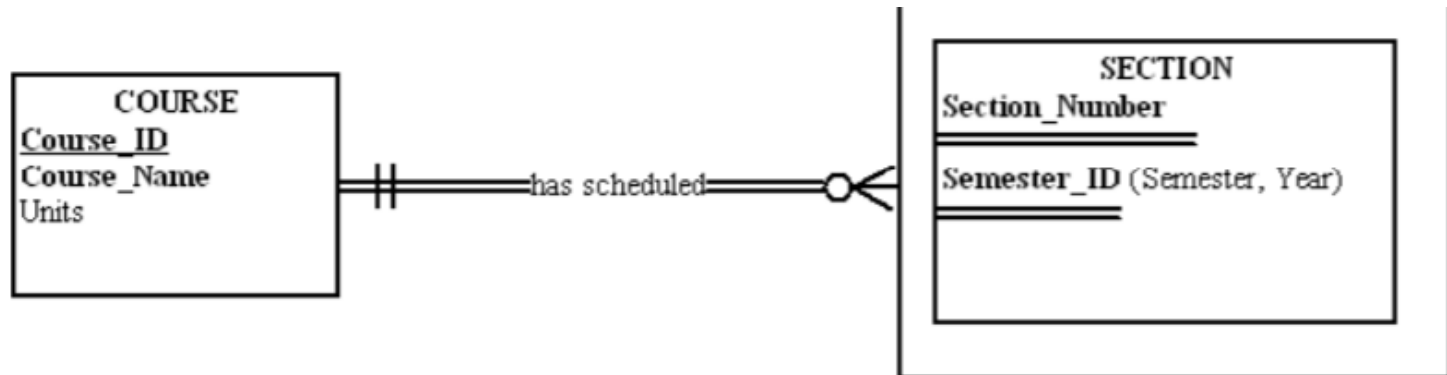
MAP WEAK ENTITIES:

- ❖ A weak entity type does not have a complete identifier but must have an attribute called a partial identifier that permits distinguishing the various occurrences of the weak entity for each owner entity instance.
- ❖ For each weak entity type, create a new relation and include all of the simple attributes
 - include the primary key of the *identifying* relation as a foreign key attribute in this new relation
- ❖ The primary key of the new relation is the combination of this primary key of the identifying and the partial identifier of the weak entity type.

MAP WEAK ENTITIES: contd..



ACTIVITY -III



MAP WEAK ENTITIES contd..

- ❖ Alternate approach to weak entity mapping
- ❖ Create a new attribute (called Dependent#), which will be used as a **surrogate primary key**
- ❖ **Surrogate primary key**
 - A serial number or other system assigned primary key for a relation.
 - DEPENDENT(Dependent#, EmployeeID, FirstName, MiddleInitial, LastName, DateOfBirth, Gender)
 - Dependent# is simply a serial number that is assigned to each dependent of an employee.

MAP WEAK ENTITIES contd..

❖ WHEN TO CREATE A SURROGATE KEY

- A surrogate key is usually created to simplify the key structures
- There is a composite primary key, as in the case of the DEPENDENT relation
- The natural primary key is inefficient
- The natural primary key is recycled

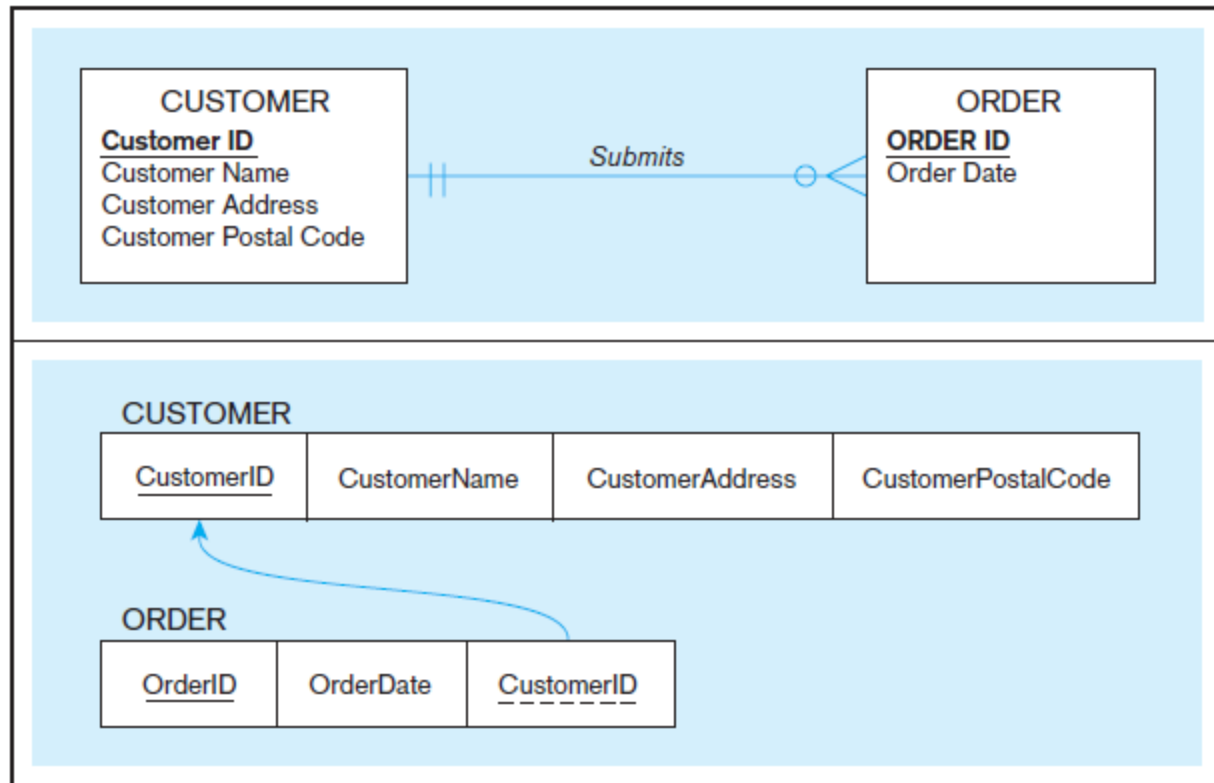
Map Binary Relationships

- ❖ Depend on the degree of relationships

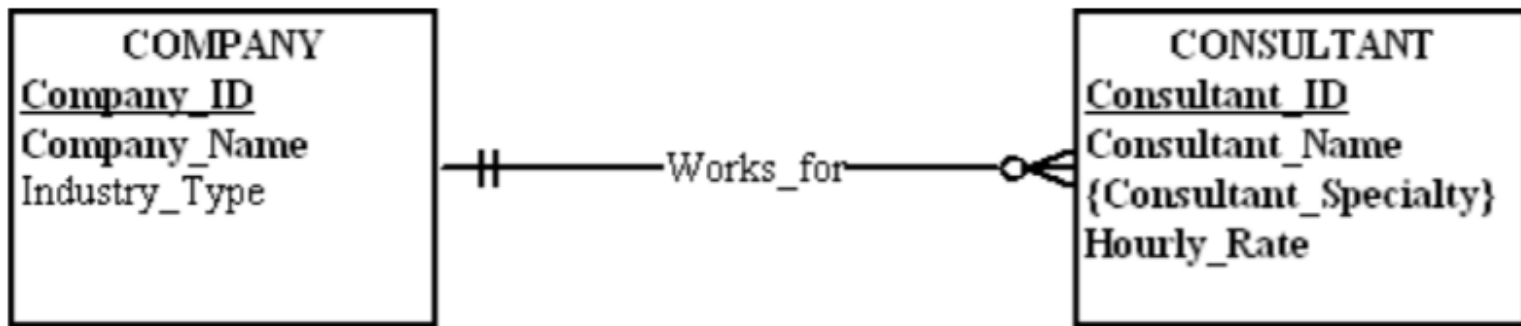
- ❖ **MAP BINARY ONE-TO-MANY RELATIONSHIPS**

- each binary 1: M relationship create
- a relation for each of the two entity types participating in the relationship
- include the primary key attribute (or attributes) of the entity on the one-side of the relationship as a foreign key in the relation that is on the many-side of the relationship

Map Binary Relationships contd..

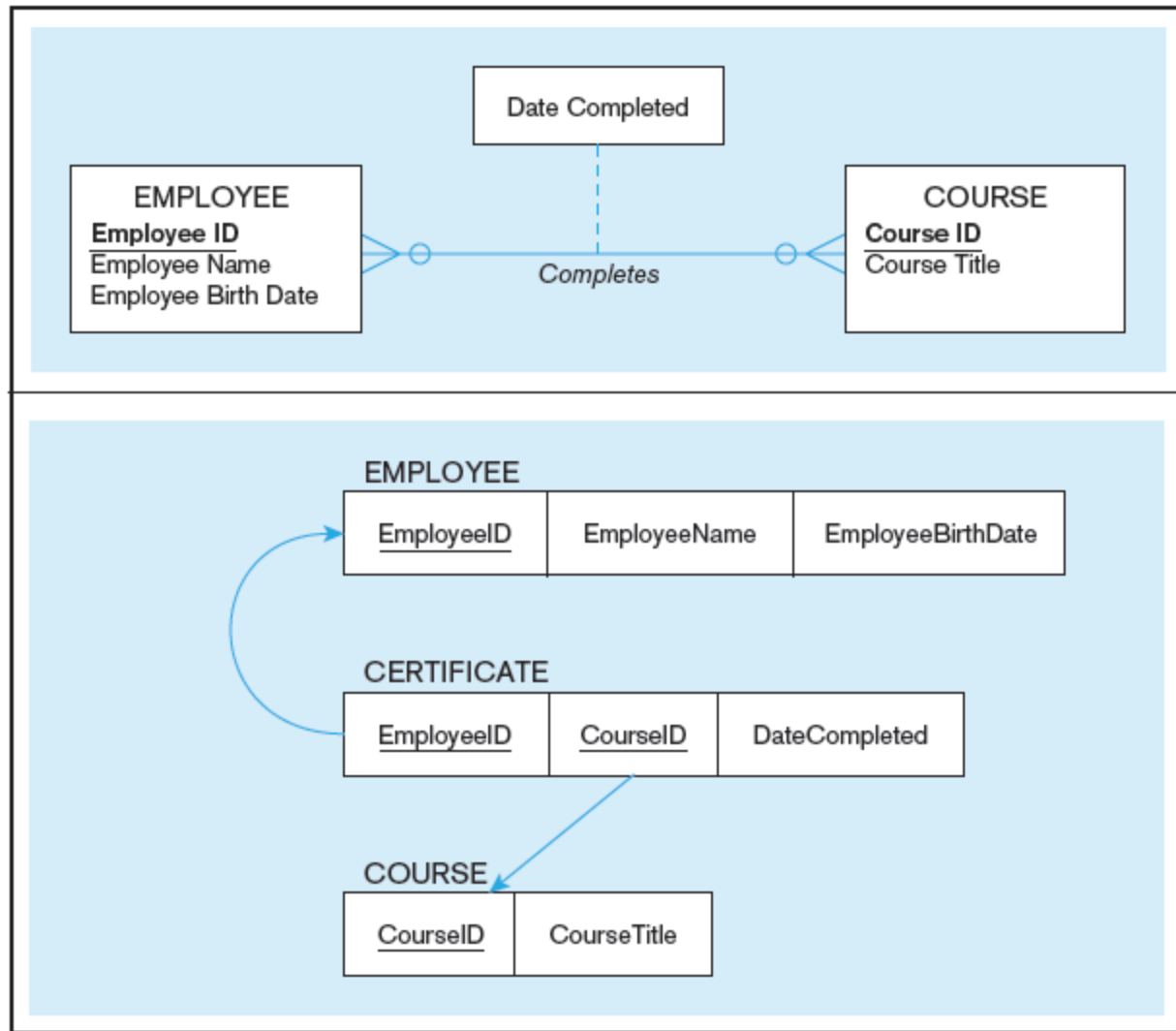


ACTIVITY-IV

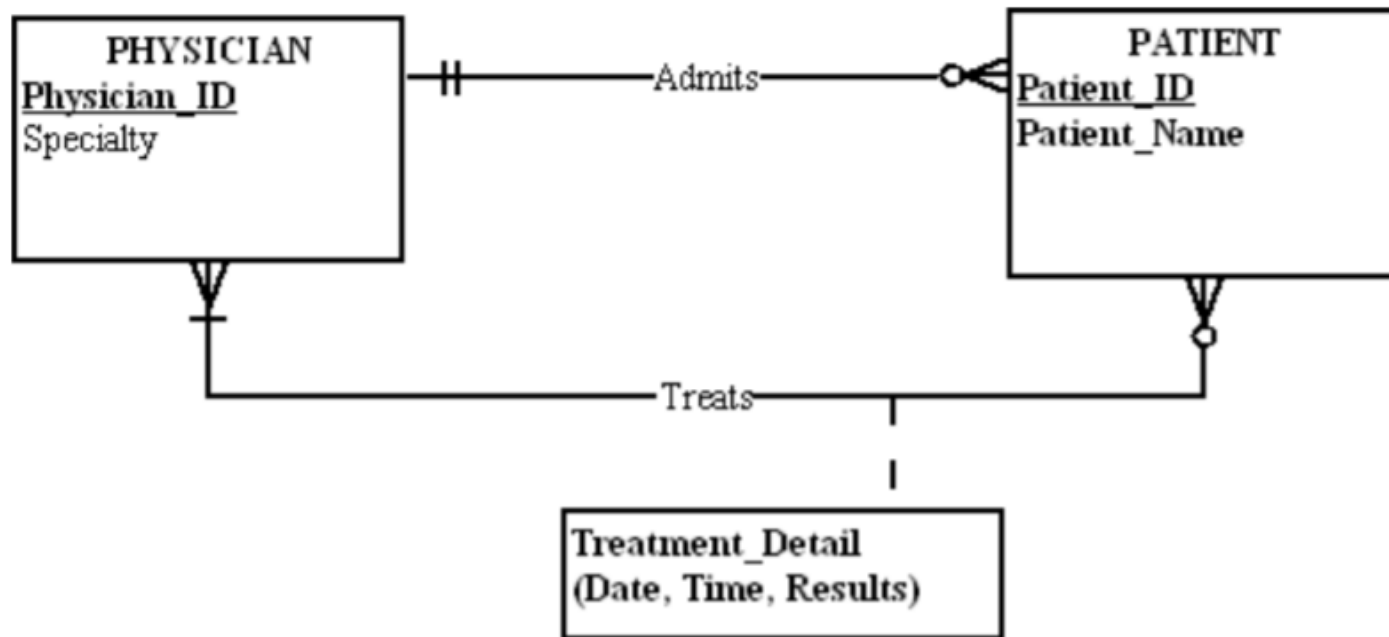


MAP BINARY MANY-TO-MANY RELATIONSHIPS

- ❖ Suppose that there is a binary many-to many ($M:N$) relationship between two entity types, A and B.
- ❖ For such a relationship, create a new relation, C
- ❖ Include as foreign key attributes in C the primary key for each of the two participating entity types
- ❖ These attributes together become the primary key of C.
- ❖ Any non key attributes that are associated with the $M:N$ relationship are included with the relation C.

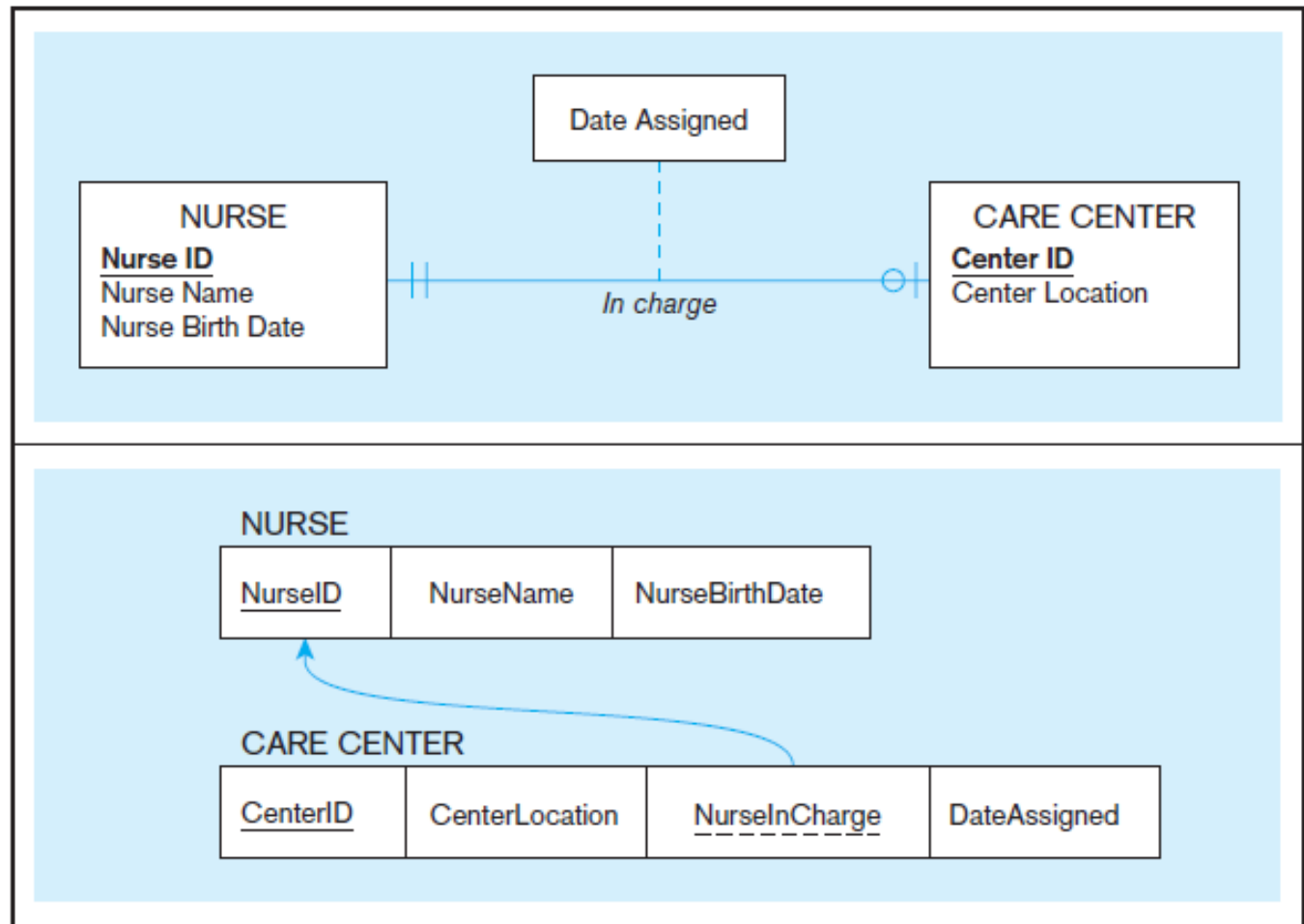


ACTIVITY-V

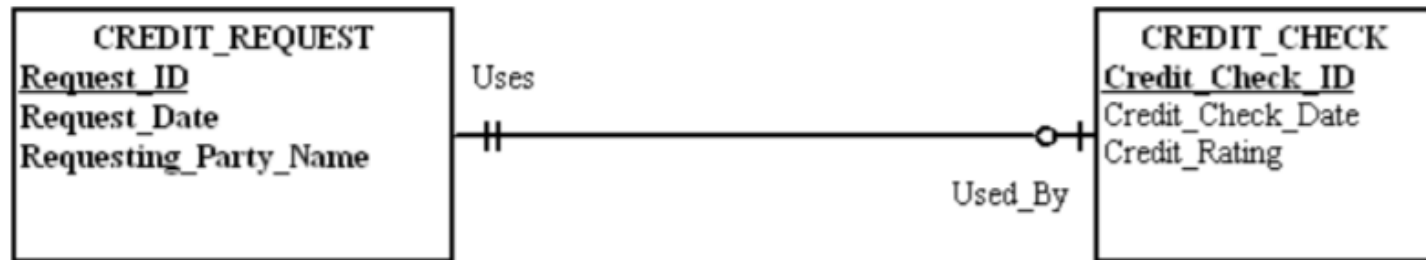


MAP BINARY ONE-TO-ONE RELATIONSHIPS

- ❖ two relations are created, one for each of the participating entity types.
- ❖ the primary key of one of the relations is included as a foreign key in the other relation.
- ❖ In a 1:1 relationship, the association in one direction is nearly always an optional one, whereas the association in the other direction is mandatory one.
- ❖ include in the relation on the optional side of the relationship the foreign key of the entity type that has the mandatory participation in the 1:1 relationship.
- ❖ Any attributes associated with the relationship itself are also included in the same relation as the foreign key.



ACTIVITY-VI



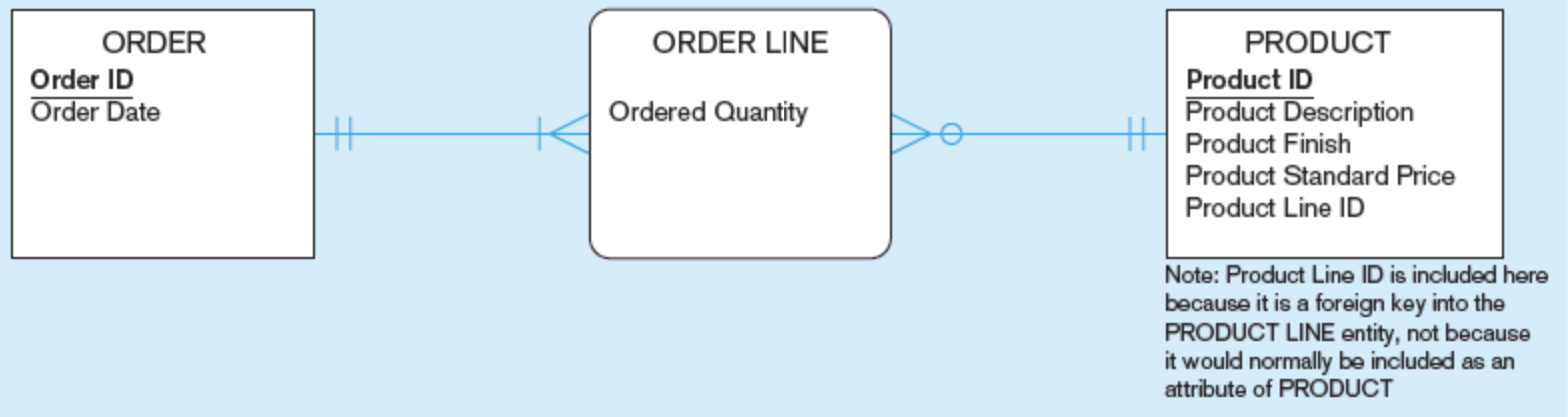
Map Associative Entities

- ❖ Mapping the associative entity involves essentially the same steps as mapping an $M:N$ relationship
- ❖ The first step is to create three relations
 - i. one for each of the two participating entity types and a third for the associative entity.
 - ii. refer to the relation formed from the associative entity as the *associative relation*.
- ❖ second step then depends on whether on the E-R diagram an identifier was assigned to the associative entity

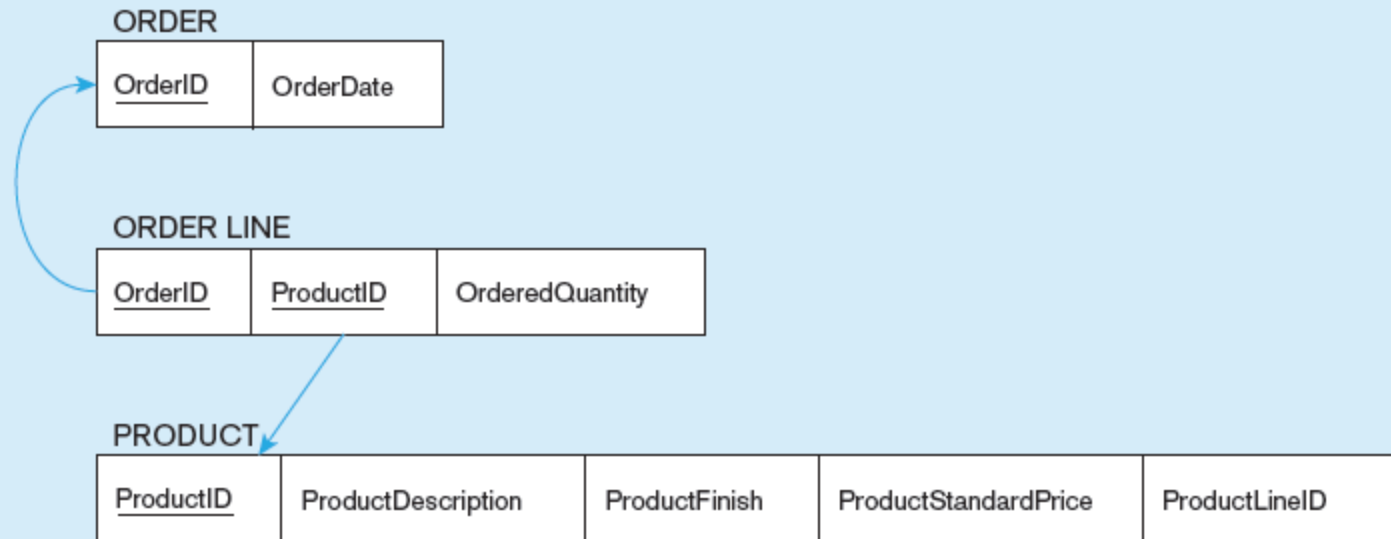
IDENTIFIER NOT ASSIGNED

- ❖ the default primary key for the associative relation consists of the two primary key attributes from the other two relations
- ❖ These attributes are then foreign keys that reference the other two relations

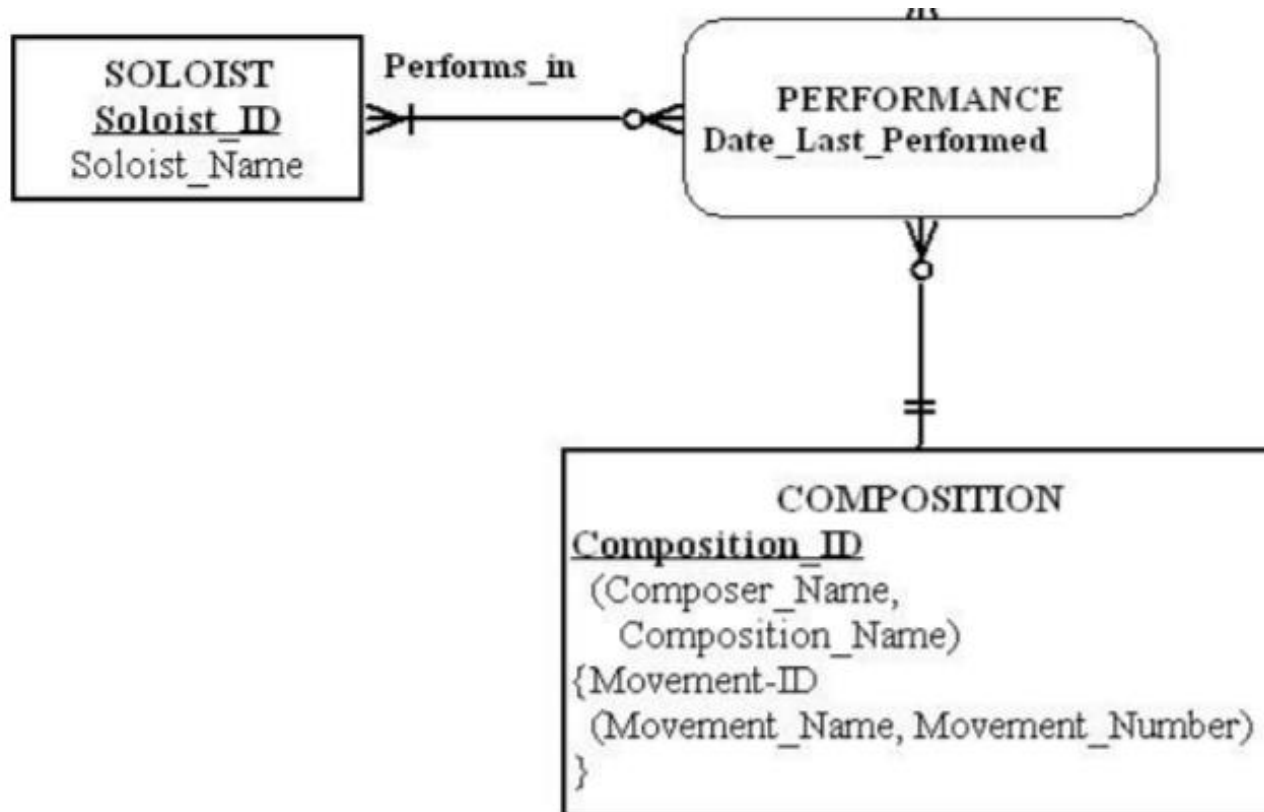
(a) An associative entity



(b) Three resulting relations

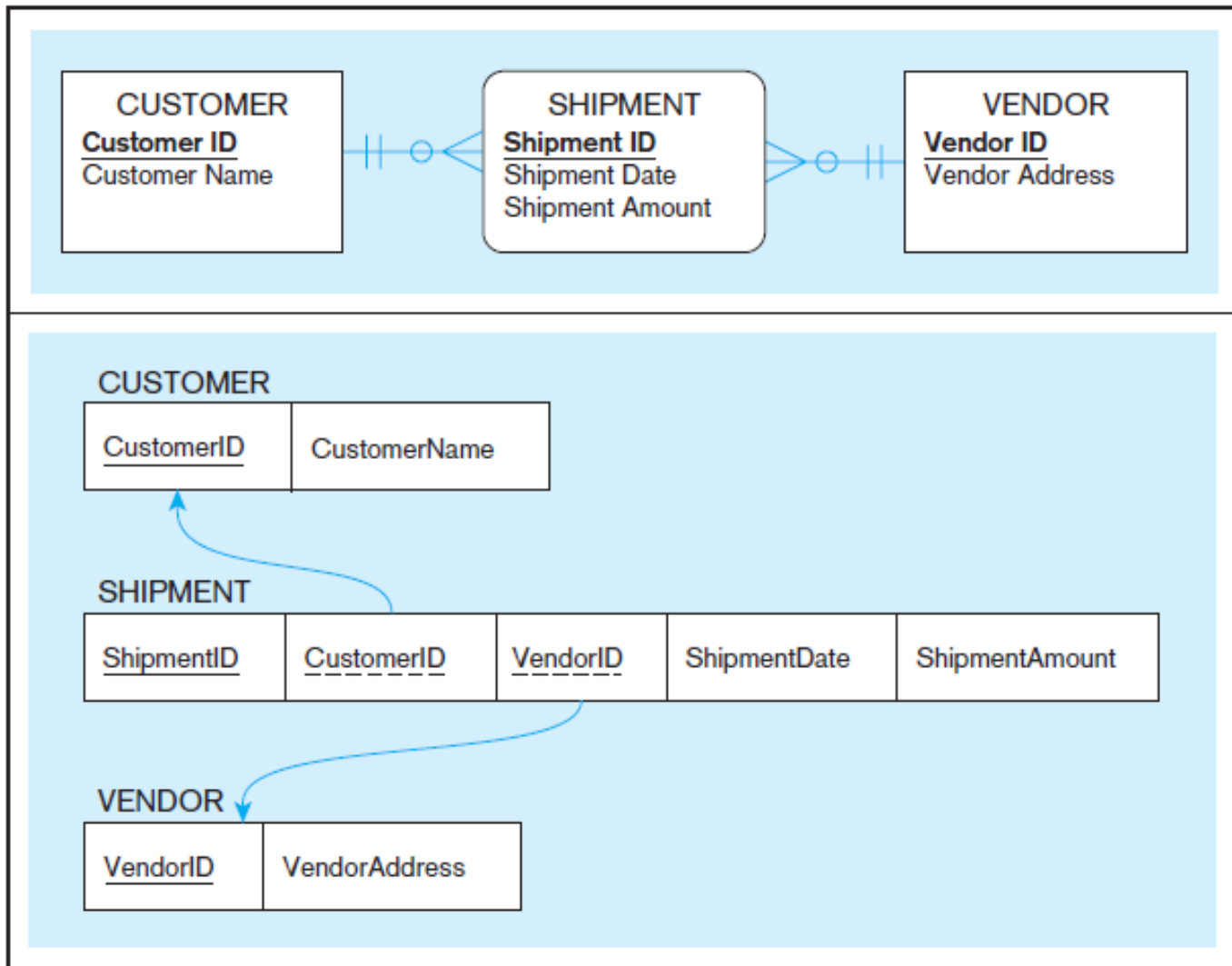


ACTIVITY-VII



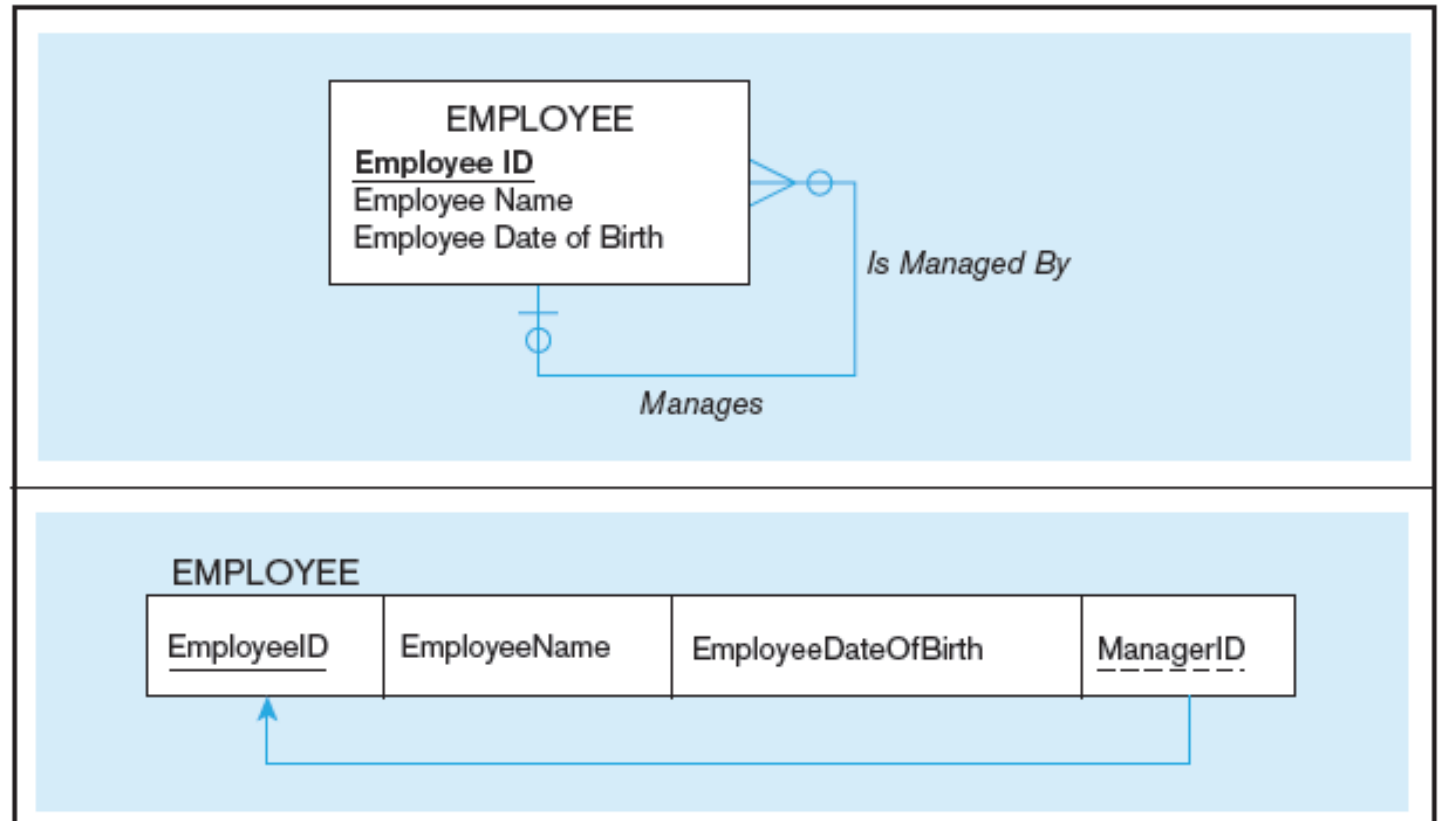
IDENTIFIER ASSIGNED

- ❖ a new (associative) relation is created to represent the associative entity.
- ❖ the primary key for this relation is the identifier assigned on the E-R diagram
- ❖ The primary keys for the two participating entity types are then included as foreign keys in the associative relation.

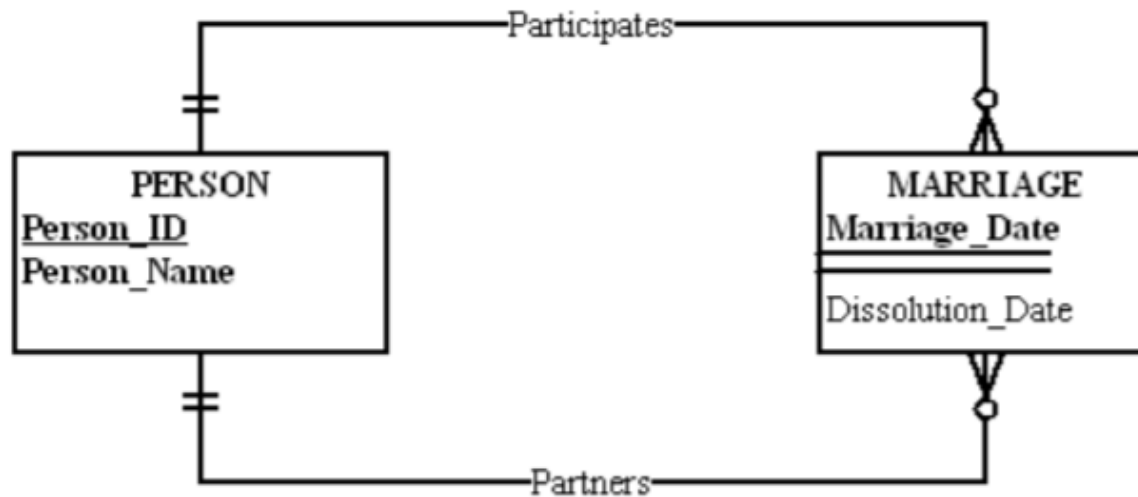


Map Unary Relationships

- ❖ also called *recursive relationships*.
- ❖ Two most important cases of unary relationships are one-to-many and many-to-many relationships.
- ❖ **UNARY ONE-TO-MANY RELATIONSHIPS**
- ❖ Map the entity type into relation
- ❖ Then a foreign key attribute is added to the same relation; this attribute references the primary key values in the same relation.
- ❖ This type of a foreign key is called a **recursive foreign key**.

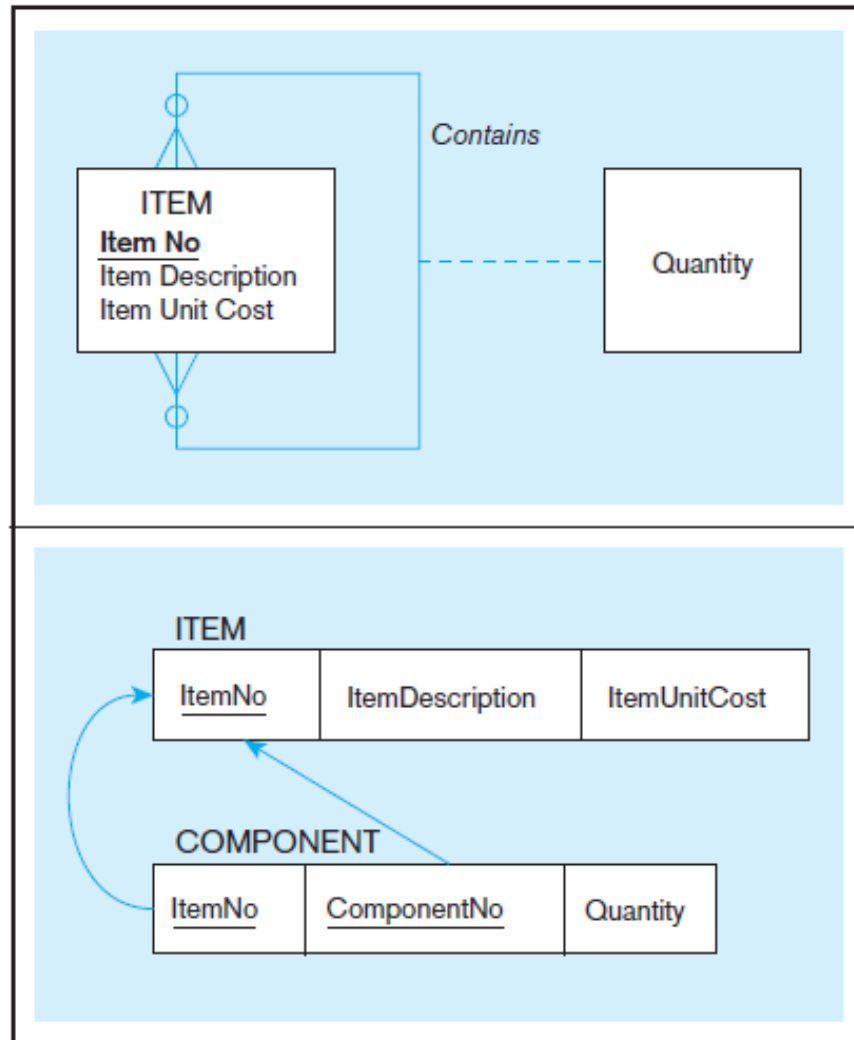


ACTIVITY-IX

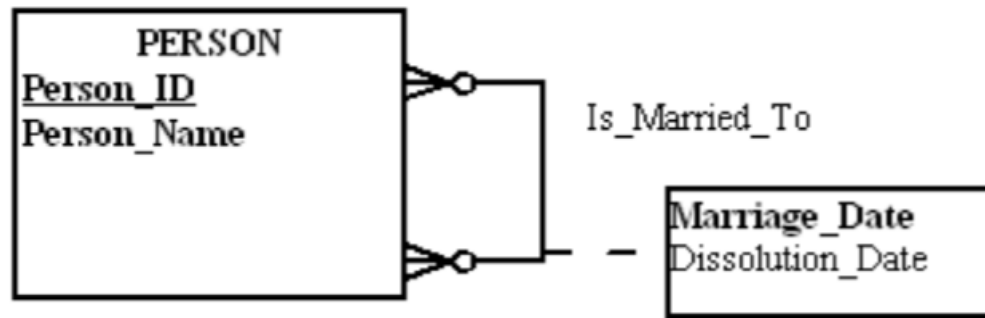


UNARY MANY-TO-MANY RELATIONSHIPS

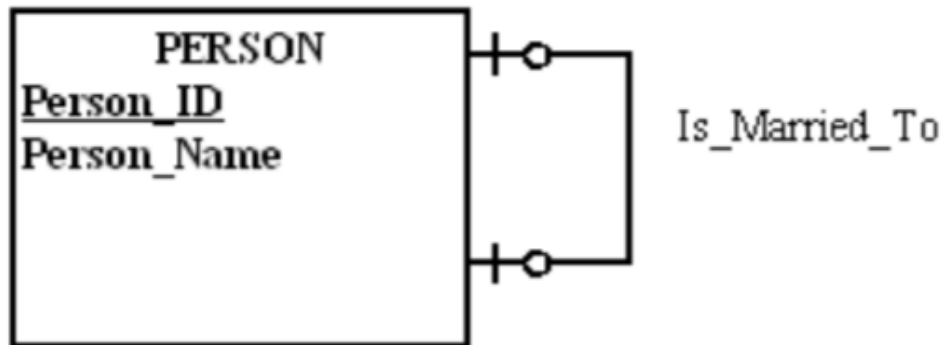
- ❖ two relations are created: one to represent the entity type in the relationship
- ❖ an associative relation to represent the $M:N$ relationship itself
- ❖ The primary key of the associative relation consists of two attributes (both attributes take their values from the primary key of the other relation).
- ❖ Any nonkey attribute of the relationship is included in the associative relation



ACTIVITY- X



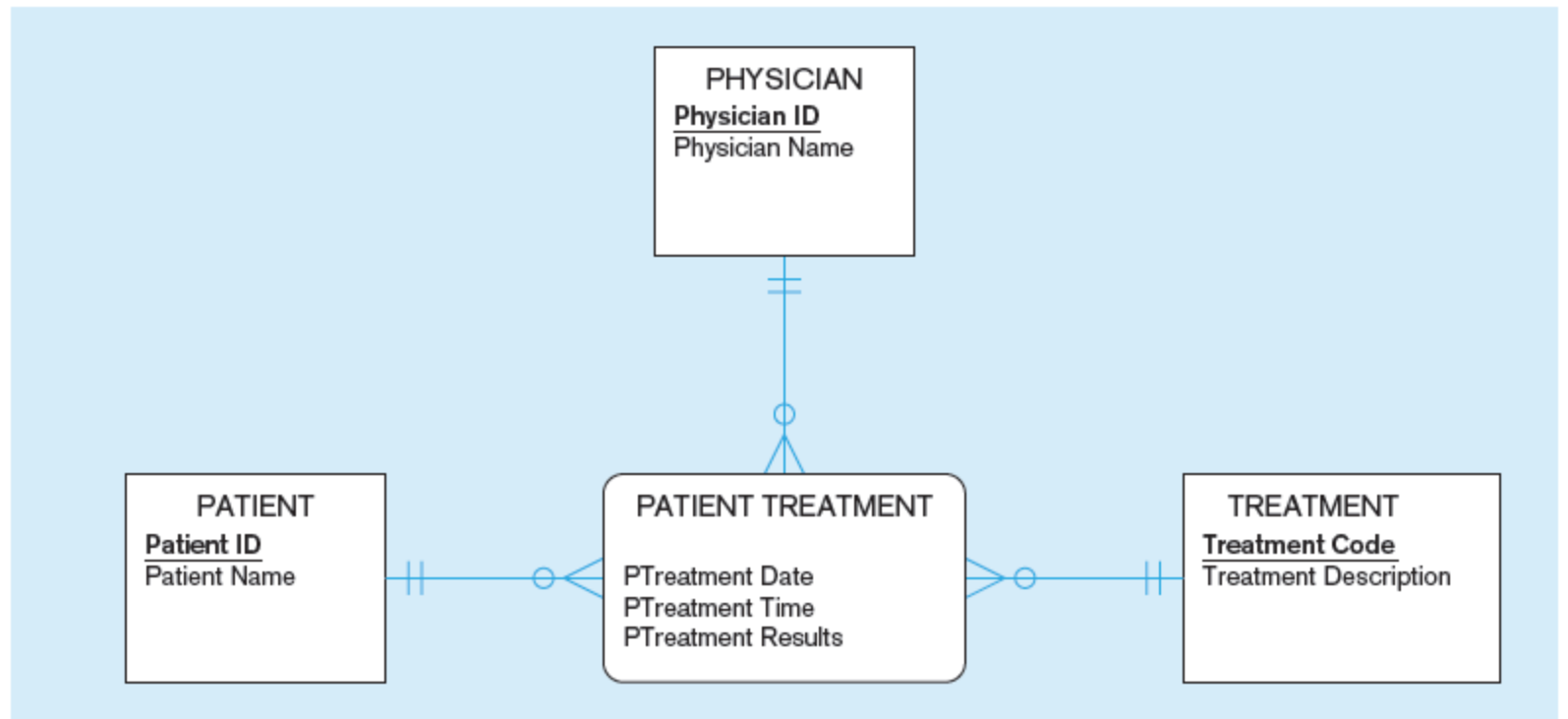
UNARY One to One



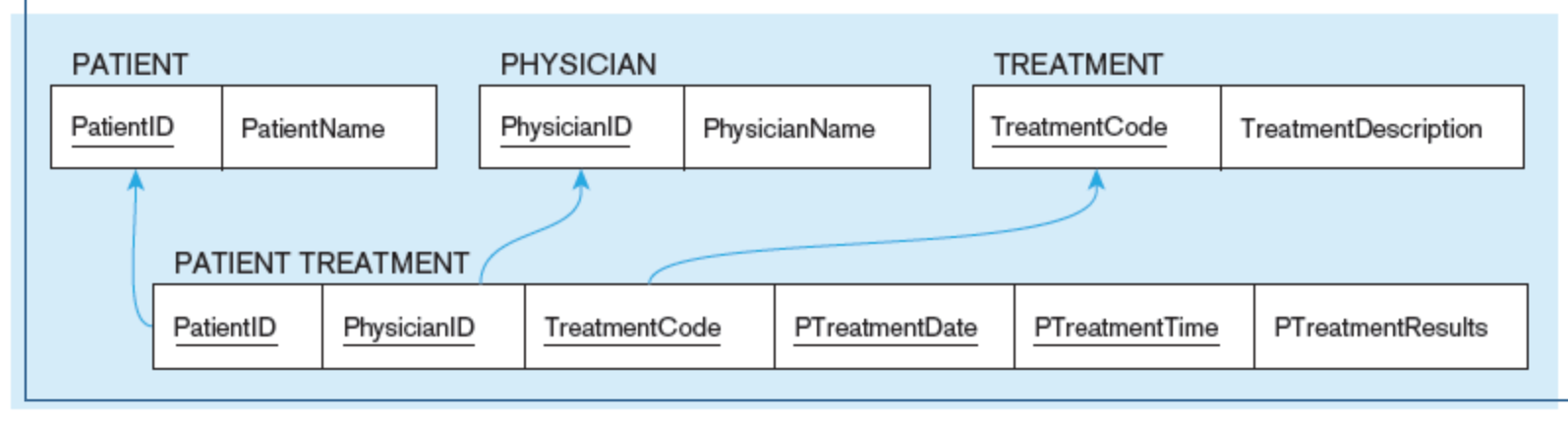
Map Ternary (and n -ary) Relationships

- ❖ It was recommended that you convert a ternary relationship to an
- ❖ associative entity to represent participation constraints more accurately.
- ❖ To map an associative entity, we create a new associative relation
- ❖ The default primary key of this relation consists of the three primary key attributes for the participating entity types.
- ❖ Any attributes of the associative entity type become attributes of the new relation.

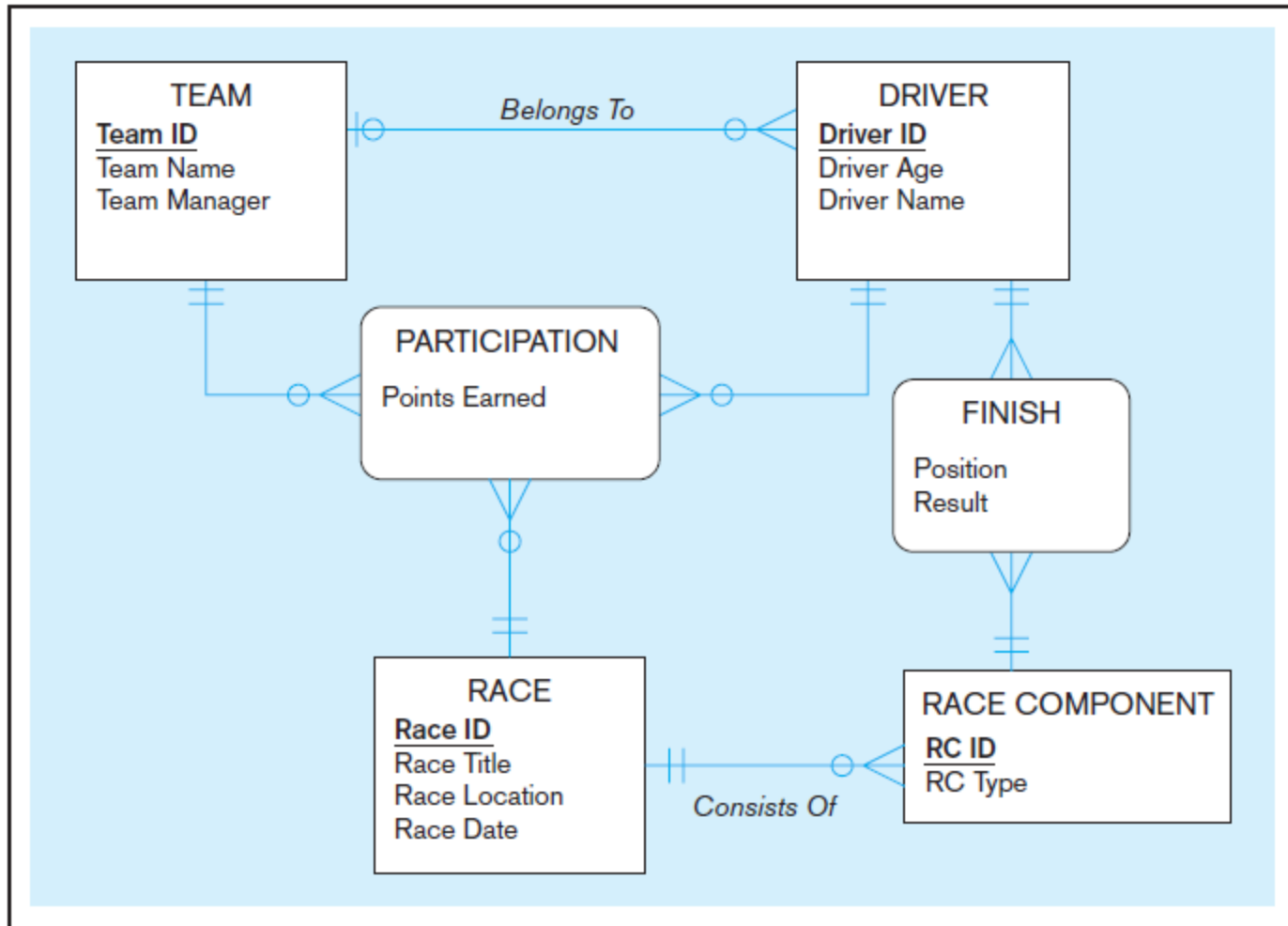
(a) PATIENT TREATMENT ternary relationship with associative entity



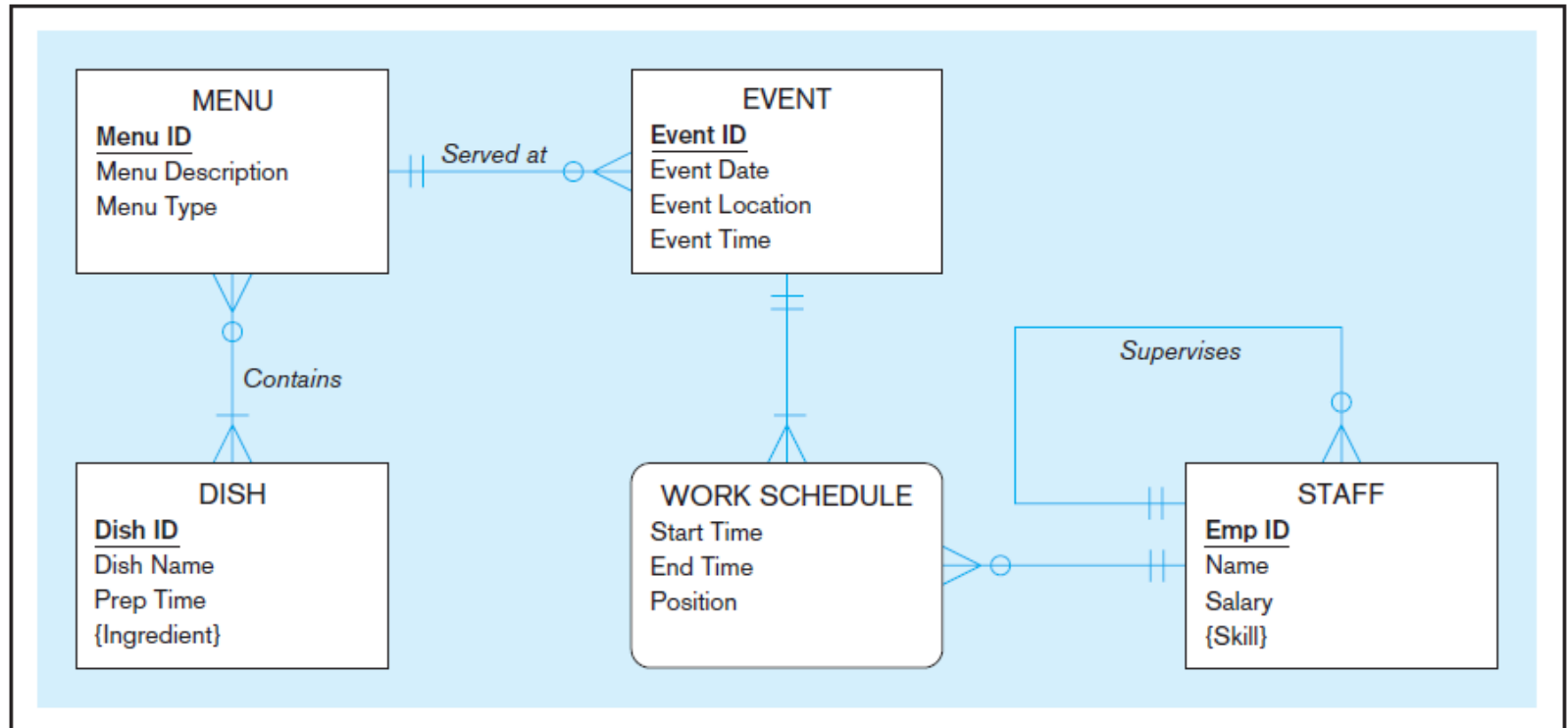
(b) Mapping the ternary relationship PATIENT TREATMENT



Activity-3

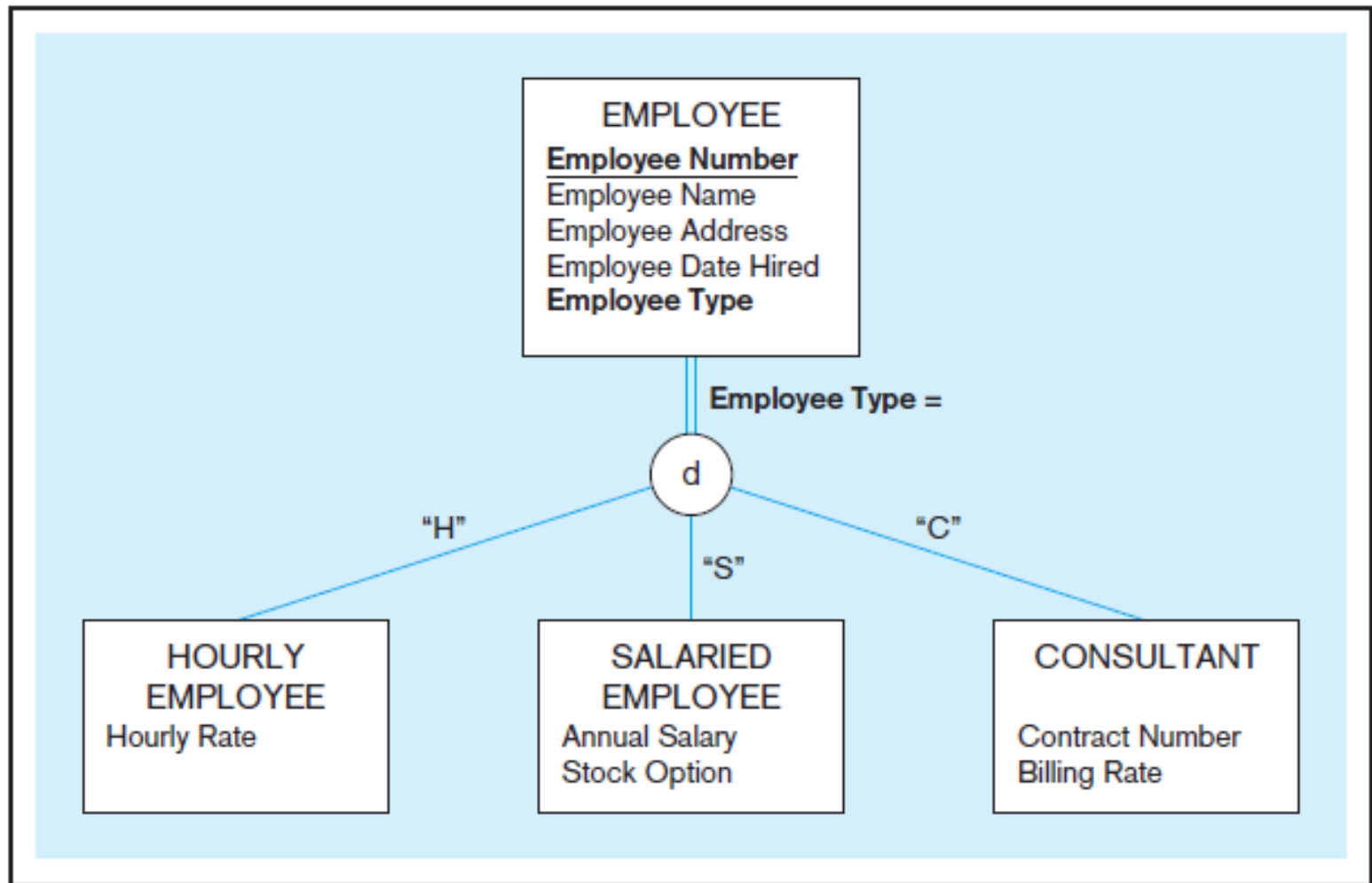


EXERCISE-2



Map Supertype/Subtype Relationships

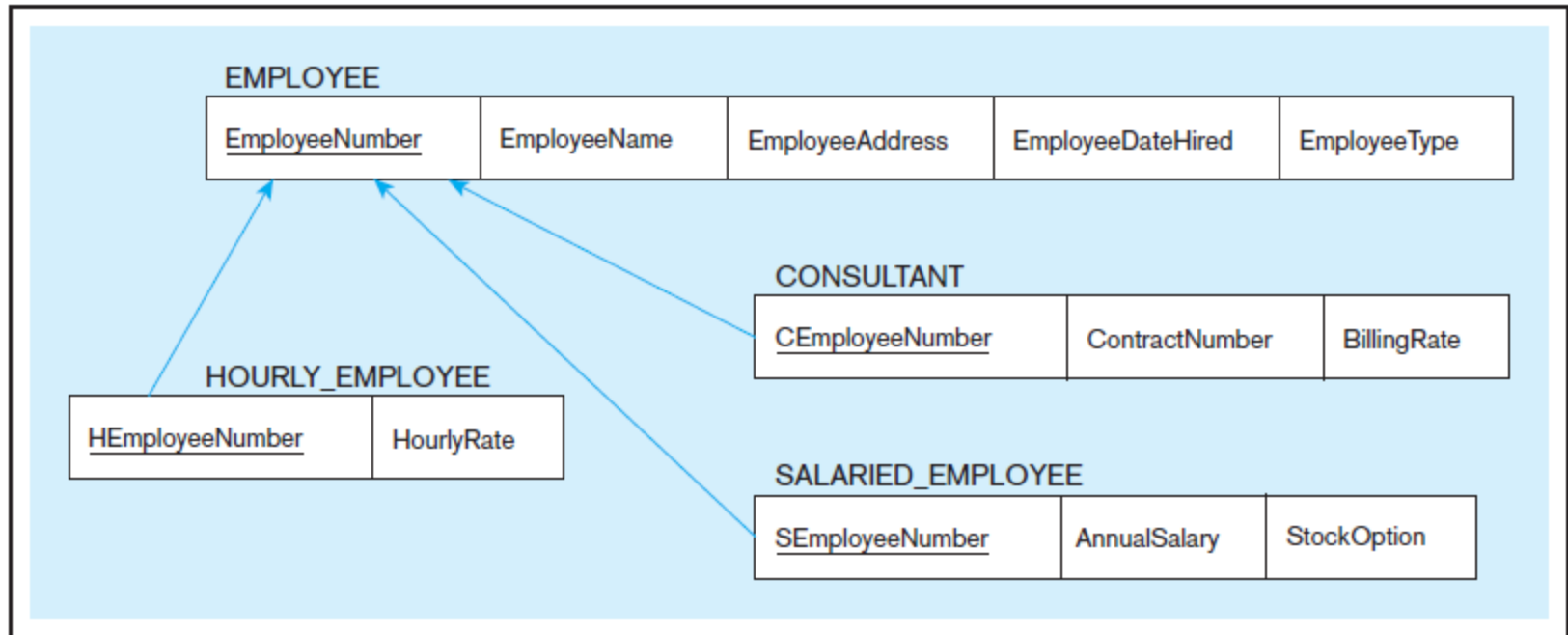
- ❖ The relational data model does not yet directly support supertype /subtype relationships
- ❖ there are various strategies that database designers can use to represent these relationships with the relational data model
- ❖ Use the following strategy, which is the one most commonly employed:
 - i. Create a separate relation for the supertype and for each of its subtypes.
 - ii. Assign to the relation created for the supertype the attributes that are common to all members of the supertype, including the primary key.
 - iii. Assign to the relation for each subtype the primary key of the supertype and only those attributes that are unique to that subtype. Assign one (or more) attributes of the supertype to function as the subtype discriminator.



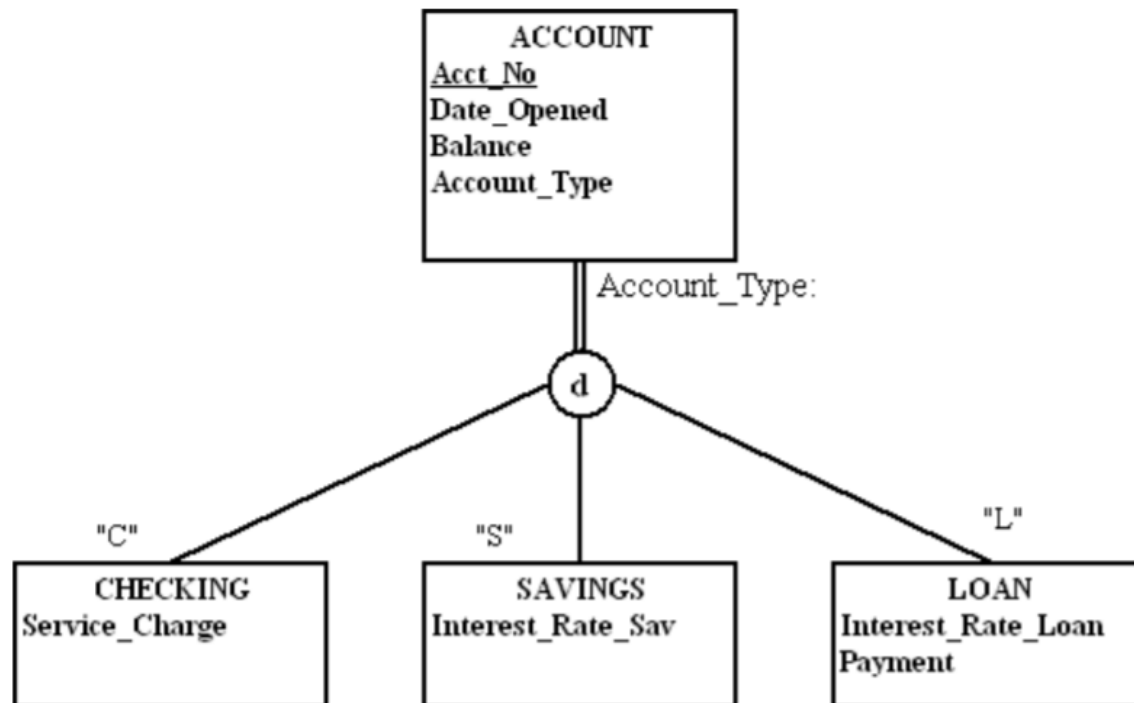
Map Supertype/Subtype Relationships contd..

- ❖ There is one relation for the supertype (EMPLOYEE) and one for each of the three subtypes
- ❖ The primary key for each of the four relations is employeeNumber
- ❖ A prefix is used to distinguish the name of the primary key for each subtype
- ❖ Each of these attributes is a foreign key that references the supertype primary key

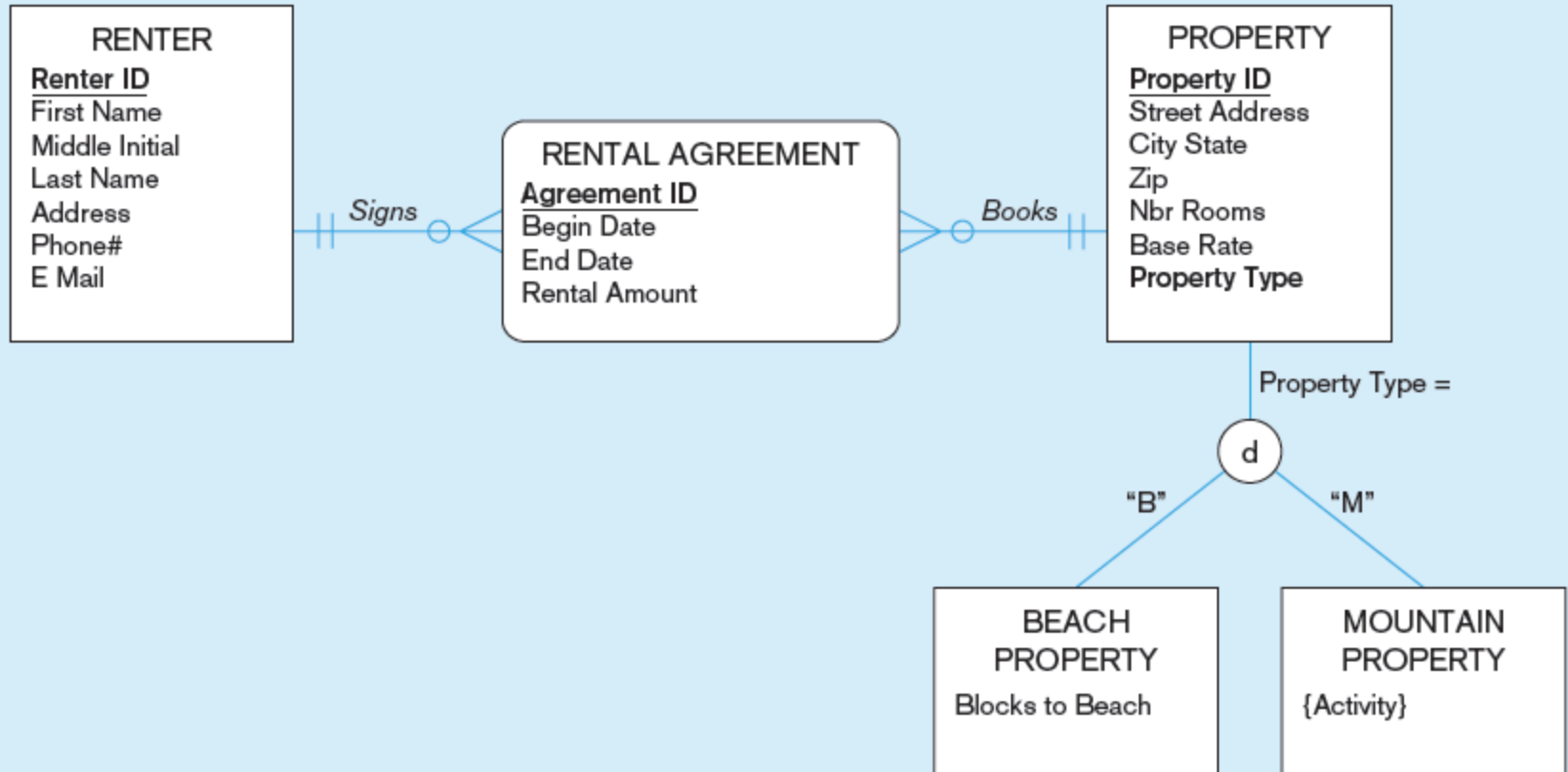
Map Supertype/Subtype Relationships contd..



ACTIVITY-XI



EXERCISE-03



Summary

- ❖ Logical database design is the process of transforming the conceptual data model into a logical data model.
- ❖ The relational data model represents data in the form of tables called relations.
- ❖ A relation is a named, two-dimensional table of data.
- ❖ A key property of relations is that they cannot contain multivalued attributes.